

HR EMPLOYEE ATTRITION

SQL & Power BI • End-to-End Project Guide

Work through every question. Build every query. Ship a portfolio-ready dashboard.

Dataset	IBM HR Analytics — Employee Attrition
Rows	1,470 employees
Columns	35 fields (demographics, job info, satisfaction, compensation)
Target	Attrition (Yes = 237 / No = 1,233) — 16.1% attrition rate
Tools	SQL (any RDBMS) + Power BI Desktop

How to use this guide. Each phase contains numbered questions. Try to answer them yourself first — write the SQL, make the chart, form the insight. The full answer sheet starts after the last phase. The questions mirror real analyst interview tasks, so completing this project end-to-end gives you both a portfolio piece and practice for technical interviews.

Table of Contents

Phase 0	Setup & Data Loading	p. 3
Phase 1	SQL — Exploratory Data Analysis	p. 4
Phase 2	SQL — Attrition Deep Dive	p. 5
Phase 3	SQL — Advanced Analysis	p. 7
Phase 4	Power BI — Data Model	p. 9
Phase 5	Power BI — DAX Measures	p. 10
Phase 6	Power BI — Dashboard Build	p. 11
Phase 7	Insights & Storytelling	p. 13
■ ■ ■	ANSWER SHEET (all phases)	p. 14

Before writing a single query, you need a clean table in your database. This phase covers creating the schema, loading the CSV, and confirming the data looks right. Use any RDBMS — PostgreSQL, MySQL, SQL Server, or SQLite all work.

Setup Questions

Q1

Create a table called `hr_attrition` with the correct column names and data types from the `HR-Employee-Attrition.csv` file. Which columns should be `VARCHAR` and which should be `INT`?

■ Open the CSV header. Categorical text fields → `VARCHAR`. Numeric fields → `INT`. Boolean-like fields (`OverTime`, `Attrition`, `Over18`) → `VARCHAR(3)`.

Q2

After loading the data, write a query to confirm all 1,470 rows loaded successfully and check for `NULL` values in the `Attrition` column.

■ Use `COUNT(*)` and `COUNT(Attrition)` — if they differ, you have `NULL`s.

Q3

Some columns in this dataset have zero variance (every row has the same value). Identify which columns they are and explain why they should be excluded from analysis.

■ Run `SELECT DISTINCT` on suspicious columns like `EmployeeCount`, `Over18`, `StandardHours`.

Q4

The `Education` and `JobLevel` columns contain integers (1–5) that map to labels. Write a query that outputs `Education` as its label (1=Below College, 2=College, 3=Bachelor, 4=Master, 5=Doctor) instead of the raw number.

■ Use a `CASE WHEN` expression.

Exploratory analysis answers basic shape-of-the-data questions: how many employees, what departments exist, what the salary range looks like. These queries are quick but essential — they catch data quality issues early and give you numbers to put in your Power BI KPI cards.

EDA Questions

Q5

What is the overall attrition rate? Return three columns: total_employees, employees_left, and attrition_rate_pct rounded to 2 decimal places.

■ Use `SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END)` for the count.

Q6

How many employees are in each Department? Return department name and count, ordered from largest to smallest.

■ Simple `GROUP BY + ORDER BY DESC`.

Q7

What is the distribution of employees across JobRole? Include the count and the percentage of total headcount for each role.

■ Use `COUNT(*) * 100.0 / SUM(COUNT(*) OVER())` for the percentage.

Q8

Summarize the MonthlyIncome column: return the minimum, maximum, average, and median salary across all employees.

■ For median, use `PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY MonthlyIncome)` in PostgreSQL/SQL Server, or a row-number trick in MySQL.

Q9

How are employees distributed across age bands: Under 30, 30–39, 40–49, 50+? Return the band label, count, and percentage of total.

■ Use `CASE WHEN` to create the band, then `GROUP BY` the band expression.

Q10

What proportion of employees work OverTime? Return the OverTime value (Yes/No), count, and percentage.

■ Same pattern as Q7.

Q11

What is the average MonthlyIncome by JobLevel (1–5)? Order by JobLevel ascending.

■ Confirm the obvious expectation: higher JobLevel = higher pay.

This is the core of the project. Every query in this phase answers a specific business question that HR leadership would actually ask. Focus on computing attrition rate (not just count) so comparisons between groups are meaningful.

Attrition by Segment

Q12

What is the attrition rate by Department? Return department, total headcount, number who left, and attrition rate %. Order by attrition rate descending.

■ This reveals which department HR should focus on first.

Q13

What is the attrition rate by JobRole? Include the same columns as Q12.

■ Expect Sales Representative and Laboratory Technician to be near the top.

Q14

Does overtime drive attrition? Compare the attrition rate for employees who work overtime vs those who don't.

■ This is likely to be one of the strongest predictors in the dataset.

Q15

Is there an attrition difference by Gender? By MaritalStatus?

■ Run two separate queries or combine with UNION ALL.

Compensation & Promotion

Q16

Compare the average MonthlyIncome of employees who left vs those who stayed. Do this at the overall level AND broken down by Department.

■ Use two CTEs — one for leavers, one for stayers — then JOIN them.

Q17

Group YearsSinceLastPromotion into buckets (0–1 years, 2–3 years, 4–6 years, 7+ years) and calculate the attrition rate for each bucket.

■ Stale promotion status is a classic retention risk signal.

Q18

Does a higher PercentSalaryHike correlate with lower attrition? Group hike % into Low ($\leq 12\%$), Medium (13–19%), High ($\geq 20\%$) and compare attrition rates.

■ Use CASE WHEN on PercentSalaryHike for bucketing.

Satisfaction Scores

Q19

For employees who left vs those who stayed, compare the average scores for: JobSatisfaction, EnvironmentSatisfaction, WorkLifeBalance, and RelationshipSatisfaction. Show all four in a single result set.

■ Use `AVG()` with a `CASE WHEN` inside, or a `FILTER` clause.

Q20

What is the attrition rate for each JobSatisfaction level (1–4)? Label the levels: 1=Low, 2=Medium, 3=High, 4=Very High.

■ `CASE WHEN` for labels, then `GROUP BY` the raw score.

Q21

Are employees with low WorkLifeBalance (score = 1) significantly more likely to leave? Compute the attrition rate for each WorkLifeBalance level.

■ Same pattern as Q20.

This phase uses intermediate-to-advanced SQL — window functions, multiple CTEs, and conditional aggregation. These queries produce the insights that make a portfolio stand out. They also map directly to the Power BI visuals in Phase 6.

Window Functions & Ranking

Q22

Using a window function, rank each JobRole by its attrition rate from highest to lowest. Return role name, attrition rate, and rank.

■ Use `RANK()` or `DENSE_RANK()` `OVER (ORDER BY attrition_rate DESC)` on a CTE.

Q23

For each Department, find the JobRole with the highest attrition rate. Use a window function — do not use a subquery or manual filtering.

■ `RANK()` `OVER (PARTITION BY Department ORDER BY attrition_rate DESC)`, then filter `WHERE rank = 1`.

Q24

Compute a running total of attrition count as YearsAtCompany increases from 0 to 40. This shows whether early-tenure employees leave the most.

■ `SUM(leavers) OVER (ORDER BY YearsAtCompany ROWS UNBOUNDED PRECEDING)`.

Multi-Factor Analysis

Q25

Write a query that computes the attrition rate for every combination of OverTime and JobSatisfaction level. Which combination has the worst attrition?

■ `GROUP BY OverTime, JobSatisfaction` — 8 combinations total (2×4).

Q26

For employees who left, what is the average TotalWorkingYears, YearsAtCompany, and YearsInCurrentRole? Compare to employees who stayed.

■ This builds the 'typical leaver profile' you'll describe in Phase 7.

Q27

Does NumCompaniesWorked (number of previous employers) relate to attrition? Group into: 0 (first job), 1–2 (few companies), 3–5 (moderate), 6+ (job hopper) and compute attrition rate per group.

■ `CASE WHEN` bucketing, then `GROUP BY`.

At-Risk Employee Identification

Q28

Build a CTE that flags each employee with a risk score. Add 1 point for each of the following conditions: `OverTime = 'Yes'`, `JobSatisfaction ≤ 2`, `WorkLifeBalance = 1`, `YearsSinceLastPromotion ≥ 4`, `MonthlyIncome` below department average. Return a count of employees at each risk level (0–5).

■ Use a CTE for the department avg salary, another for the flags, then SUM the flags as risk_score.

Q29

From the risk score CTE in Q28, list the `EmployeeNumber`, `Department`, `JobRole`, and `risk_score` for all employees with a score of 4 or 5. These are your 'high risk' employees. How many are there? 28

■ Add `WHERE risk_score ≥ 4` to your Q28 CTE output.

Q30

Validate your risk model: for employees who actually left (`Attrition = 'Yes'`), what was their average risk score? For those who stayed? A higher score for leavers means the model is directionally correct.

■ Join the risk CTE back to the base table and AVG the score by Attrition.

A clean data model makes DAX easy and slicers fast. For this project the model is simple — one fact table and a few helper tables — but getting it right matters.

Data Model Questions

Q31

Import the CSV into Power BI via Power Query. In Power Query, what transformations should you apply before closing & loading? List at least four.

■ *Think: column types, removing zero-variance columns, renaming columns for readability, adding a satisfaction label column.*

Q32

Create a calculated column in Power Query (or DAX) called AttritionFlag that returns 1 if Attrition = 'Yes' and 0 if 'No'. Why is this column useful?

■ *It allows you to SUM() instead of COUNTIF — much easier in DAX.*

Q33

Add a calculated column called AgeGroup that buckets Age into: Under 30 / 30–39 / 40–49 / 50+. Write the DAX.

■ *Use SWITCH(TRUE(), Age < 30, ...) or nested IF.*

Q34

Add a calculated column called TenureGroup that buckets YearsAtCompany into: 0–2 years / 3–5 years / 6–10 years / 10+ years.

■ *Same SWITCH(TRUE()) pattern.*

Q35

Should you create a separate Date table for this dataset? Explain your reasoning. If yes, what columns would it need?

■ *Hint: there are no date columns in this dataset — only year/tenure integers.*

All key metrics should be explicit measures — not implicit aggregations. This gives you control over formatting, context transitions, and filter behavior. Write every measure below in a dedicated 'Key Measures' table.

Core Measures

Q36

Write DAX measures for: Total Employees, Total Attrition (count of leavers), and Attrition Rate %.

■ Use `SUM([AttritionFlag])` for attrition count. `DIVIDE()` for the rate to avoid division-by-zero errors.

Q37

Write a measure for Avg Monthly Income. Then write a second measure called Avg Income — Leavers that calculates average income only for employees who left.

■ Use `CALCULATE(AVERAGE(...), hr_attrition[Attrition] = "Yes")`.

Q38

Write a measure called Avg Job Satisfaction — Leavers and another called Avg Job Satisfaction — Stayers. These will be used in a side-by-side bar chart.

■ `CALCULATE(AVERAGE([JobSatisfaction]), [Attrition] = "Yes/No")`.

Dynamic & Context-Aware Measures

Q39

Write a measure called % of Total Employees that returns the percentage of employees represented by the current filter context. It should work correctly inside a bar chart broken down by Department.

■ `DIVIDE([Total Employees], CALCULATE([Total Employees], ALL(hr_attrition)))`.

Q40

Write a measure called High Risk Employee Count that counts employees who work overtime AND have JobSatisfaction ≤ 2. Format it as a KPI card.

■ `CALCULATE(COUNTROWS(hr_attrition), [OverTime]="Yes", [JobSatisfaction] <= 2)`.

Q41

Write a measure called Attrition Rate — Selected Dept that returns the attrition rate for whichever department is selected in a slicer, and returns the overall rate when nothing is selected.

■ `CALCULATE([Attrition Rate %])` with no extra filter — filter context flows from the slicer automatically.

Build three report pages. Every visual must have a clear business question it answers — no decoration. Use consistent colors: one color for 'Attrition = Yes', another for 'No'. Add slicers on every page so the report is interactive.

Page 1 — Executive Overview

Q42

Add four KPI cards to Page 1: Total Employees, Total Attrition, Attrition Rate %, and Avg Monthly Income. What format should the Attrition Rate % card use?

■ Percentage format with 1 decimal place. Use conditional formatting to color the card red if rate > 15%.

Q43

Add a Donut Chart showing the split between Attrition Yes and No. Which field goes in the Legend slot and which in the Values slot?

■ Legend = Attrition, Values = Total Employees (or count). Label the donut with percentages, not raw counts.

Q44

Add a Clustered Bar Chart showing Attrition Rate % by Department. Sort it by attrition rate descending. What insight should this chart surface immediately?

■ Sort by the measure value, not alphabetically. The highest-attrition department should be instantly visible.

Q45

Add a Stacked Bar Chart showing headcount by Department broken down by Attrition. Why is this chart useful alongside the attrition rate bar from Q44?

■ Rate shows proportion; stacked bar shows absolute scale — a small dept with 50% attrition may be less urgent than a large one at 20%.

Q46

Add slicers for Department, Gender, and OverTime on Page 1. What slicer style is best for a field with only 2–3 values?

■ Dropdown for many values, Tile/Button style for 2–4 values like Gender or OverTime.

Page 2 — Employee Deep Dive

Q47

Add a Scatter Plot with MonthlyIncome on the X-axis, TotalWorkingYears on the Y-axis, and color by Attrition. What pattern do you expect to see?

■ Leavers should cluster toward lower income and fewer working years.

Q48

Add a Bar Chart showing Attrition Rate % by JobRole, sorted descending. Add data labels showing the exact percentage. Which role has the highest rate?

■ Expected: Sales Representative or Laboratory Technician near the top.

Q49

Add a Matrix visual with JobLevel (rows) and OverTime (columns) showing Attrition Rate % in each cell. Apply conditional formatting — darker = higher rate.

■ *This shows the intersection of two risk factors simultaneously.*

Q50

Add a Histogram (use a Clustered Column chart) showing Age distribution for leavers vs stayers using the AgeGroup calculated column. What age group leaves the most?

■ *Use AgeGroup on X-axis, Total Attrition and Total Employees as Values, or filter to leavers only using the AttritionFlag.*

Page 3 — Satisfaction & Risk

Q51

Add a Clustered Bar Chart comparing average satisfaction scores (JobSatisfaction, EnvironmentSatisfaction, WorkLifeBalance, RelationshipSatisfaction) for leavers vs stayers side by side. How do you display four measures on one chart?

■ *Create a summary table with unpivoted data, or use four separate measures as multiple Value fields in a single bar chart.*

Q52

Add a 100% Stacked Bar Chart showing the attrition breakdown by YearsSinceLastPromotion bucket. What does a long bar of 'Yes' indicate?

■ *Use the TenureGroup or a promo bucket calculated column. Long Yes bar = high proportion left in that group.*

Q53

Add a KPI card for High Risk Employee Count (from Q40). Below it, add a Table visual listing those high-risk employees (EmployeeNumber, Department, JobRole, OverTime, JobSatisfaction). How do you filter the table to high-risk only?

■ *Add AttritionFlag = 0 as a visual-level filter so the table only shows current employees (not already-left ones) with the risk conditions.*

Q54

Add a Bar Chart showing Attrition Rate % by OverTime (Yes/No). How large is the gap between overtime and non-overtime attrition? Format the bars in a color that communicates urgency.

■ *Expect a very large gap — this is the single strongest predictor.*

Data analysts don't just produce charts — they produce decisions. This phase asks you to synthesize everything you found and write the kind of summary you'd present to a hiring manager or HR director. Write your answers in plain language, not SQL or DAX.

Insight Questions

Q55

In one paragraph, describe the 'typical leaver' profile at this company. Use at least five attributes from the dataset.

■ Reference: age, role, overtime status, income vs avg, satisfaction scores, tenure, years since promotion.

Q56

What is the single strongest predictor of attrition in this dataset, based on your SQL analysis? Support your answer with a specific attrition rate comparison.

■ OverTime is the expected answer — employees working OT leave at roughly 30–31% vs ~10% for those who don't.

Q57

Which department should HR prioritize for a retention intervention? Justify using both attrition rate AND absolute headcount impact.

■ Rate alone misleads — a 50% attrition dept with 10 people may be less costly than 20% attrition in a 500-person dept.

Q58

Write one actionable recommendation for each of the following HR levers: (a) compensation, (b) promotion cadence, (c) overtime policy, (d) satisfaction programs.

■ Each recommendation should be specific and tied to a number from your analysis.

Q59

What are two limitations of this dataset that could affect the conclusions? What additional data would make the analysis more reliable?

■ Think: no exit interview data, no time dimension, no team/manager quality data, no data on total comp (bonus, equity).

Q60

If you had to present this project in a 5-minute slot to a non-technical HR manager, which three visuals from your Power BI dashboard would you show first and why?

■ Think about what a non-analyst cares about: 'how bad is it, who is affected, what can we do.'

❖ ANSWER SHEET ❖

Answers are below. Use them to check your work — not to skip it. The SQL shown targets standard SQL (works in PostgreSQL, MySQL 8+, SQL Server). Adjust syntax for your specific database if needed.

PHASE 0

Setup — Answers

ANSWER — Q1

Text fields → VARCHAR: Attrition, BusinessTravel, Department, EducationField, EmployeeCount, Gender, JobRole, MaritalStatus, Over18, OverTime. Numeric fields → INT: Age, DailyRate, DistanceFromHome, Education, EmployeeNumber, EnvironmentSatisfaction, HourlyRate, JobInvolvement, JobLevel, JobSatisfaction, MonthlyIncome, MonthlyRate, NumCompaniesWorked, PercentSalaryHike, PerformanceRating, RelationshipSatisfaction, StandardHours, StockOptionLevel, TotalWorkingYears, TrainingTimesLastYear, WorkLifeBalance, YearsAtCompany, YearsInCurrentRole, YearsSinceLastPromotion, YearsWithCurrManager.

```
CREATE TABLE hr_attrition (  
  EmployeeNumber      INT PRIMARY KEY,  
  Age                 INT,  
  Attrition            VARCHAR(3),  
  BusinessTravel       VARCHAR(50),  
  DailyRate           INT,  
  Department           VARCHAR(50),  
  DistanceFromHome     INT,  
  Education            INT,  
  EducationField       VARCHAR(50),  
  EnvironmentSatisfaction INT,  
  Gender               VARCHAR(10),  
  HourlyRate           INT,  
  JobInvolvement       INT,  
  JobLevel             INT,  
  JobRole              VARCHAR(50),  
  JobSatisfaction       INT,  
  MaritalStatus        VARCHAR(20),  
  MonthlyIncome        INT,  
  NumCompaniesWorked   INT,  
  OverTime             VARCHAR(3),  
  PercentSalaryHike    INT,  
  PerformanceRating    INT,  
  RelationshipSatisfaction INT,  
  StockOptionLevel     INT,  
  TotalWorkingYears    INT,  
  TrainingTimesLastYear INT,  
  WorkLifeBalance      INT,  
  YearsAtCompany       INT,  
  YearsInCurrentRole   INT,  
  YearsSinceLastPromotion INT,  
  YearsWithCurrManager INT  
);
```

ANSWER — Q2

Both counts should return 1470. If COUNT(Attrition) < 1470, there are NULLs.

```
SELECT COUNT(*) AS total_rows,  
       COUNT(Attrition) AS non_null_attrition  
FROM hr_attrition;  
  
-- Check distinct values while you're at it:  
SELECT DISTINCT Attrition FROM hr_attrition;
```

ANSWER — Q3

Three columns have zero variance: EmployeeCount (always 1), Over18 (always 'Y'), StandardHours (always 80). They add no predictive value and should be dropped or ignored.

```
SELECT COUNT(DISTINCT EmployeeCount) AS ec_unique,  
       COUNT(DISTINCT Over18)        AS o18_unique,  
       COUNT(DISTINCT StandardHours) AS sh_unique  
FROM hr_attrition;  
  
-- All return 1 – zero variance confirmed.
```

ANSWER — Q4

CASE WHEN in the SELECT clause converts the integer to its label.

```
SELECT EmployeeNumber,  
       CASE Education  
         WHEN 1 THEN 'Below College'  
         WHEN 2 THEN 'College'  
         WHEN 3 THEN 'Bachelor'  
         WHEN 4 THEN 'Master'  
         WHEN 5 THEN 'Doctor'  
       END AS EducationLabel  
FROM hr_attrition;
```


PHASE 1

EDA — Answers

ANSWER — Q5

Overall attrition rate is 16.1% (237 out of 1,470).

```
SELECT
    COUNT(*) AS total_employees,
    SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS employees_left,
    ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END)
        / COUNT(*), 2) AS attrition_rate_pct
FROM hr_attrition;
```

ANSWER — Q6

R&D; is the largest department (~65% of headcount), followed by Sales, then HR.

```
SELECT Department, COUNT(*) AS headcount
FROM hr_attrition
GROUP BY Department
ORDER BY headcount DESC;
```

ANSWER — Q7

Use a window function to avoid a subquery for the percentage.

```
SELECT
    JobRole,
    COUNT(*) AS headcount,
    ROUND(100.0 * COUNT(*) / SUM(COUNT(*)) OVER(), 1) AS pct_of_total
FROM hr_attrition
GROUP BY JobRole
ORDER BY headcount DESC;
```

ANSWER — Q8

Min ≈ \$1,009, Max ≈ \$19,999, Avg ≈ \$6,503, Median ≈ \$4,919 (right-skewed distribution).

```
-- PostgreSQL / SQL Server syntax:
SELECT
    MIN(MonthlyIncome) AS min_income,
    MAX(MonthlyIncome) AS max_income,
    ROUND(AVG(MonthlyIncome::NUMERIC), 0) AS avg_income,
    PERCENTILE_CONT(0.5) WITHIN GROUP
        (ORDER BY MonthlyIncome) AS median_income
FROM hr_attrition;
```

ANSWER — Q9

The 30–39 band is the largest. 'Under 30' has a notable attrition rate.

```
SELECT
    CASE
```

```

    WHEN Age < 30          THEN 'Under 30'
    WHEN Age BETWEEN 30 AND 39 THEN '30-39'
    WHEN Age BETWEEN 40 AND 49 THEN '40-49'
    ELSE '50+'
END AS age_band,
COUNT(*) AS headcount,
ROUND(100.0 * COUNT(*) / SUM(COUNT(*)) OVER(), 1) AS pct
FROM hr_attrition
GROUP BY age_band
ORDER BY MIN(Age);

```

ANSWER — Q10

~28% of employees work overtime. This group will have a dramatically higher attrition rate (Phase 2).

```

SELECT OverTime,
    COUNT(*) AS headcount,
    ROUND(100.0 * COUNT(*) / SUM(COUNT(*)) OVER(), 1) AS pct
FROM hr_attrition
GROUP BY OverTime;

```

ANSWER — Q11

Salary scales predictably: L1 ≈ \$2,789, L2 ≈ \$4,099, L3 ≈ \$7,920, L4 ≈ \$13,444, L5 ≈ \$17,828.

```

SELECT JobLevel,
    ROUND(AVG(MonthlyIncome), 0) AS avg_income
FROM hr_attrition
GROUP BY JobLevel
ORDER BY JobLevel;

```

PHASE 2

Attrition Deep Dive — Answers

ANSWER — Q12

HR department has the highest attrition rate (~26%), followed by Sales (~21%). R&D; is lowest (~14%). HR is a small department so the absolute impact is limited.

```
SELECT Department,
       COUNT(*) AS headcount,
       SUM(CASE WHEN Attrition='Yes' THEN 1 ELSE 0 END) AS leavers,
       ROUND(100.0 * SUM(CASE WHEN Attrition='Yes' THEN 1 ELSE 0 END)
            / COUNT(*), 1) AS attrition_rate_pct
FROM hr_attrition
GROUP BY Department
ORDER BY attrition_rate_pct DESC;
```

ANSWER — Q13

Sales Representatives have ~40% attrition, Human Resources ~23%, Laboratory Technicians ~24%. Research Directors have the lowest rate (~2.5%).

```
SELECT JobRole,
       COUNT(*) AS headcount,
       SUM(CASE WHEN Attrition='Yes' THEN 1 ELSE 0 END) AS leavers,
       ROUND(100.0 * SUM(CASE WHEN Attrition='Yes' THEN 1 ELSE 0 END)
            / COUNT(*), 1) AS attrition_rate_pct
FROM hr_attrition
GROUP BY JobRole
ORDER BY attrition_rate_pct DESC;
```

ANSWER — Q14

OverTime = Yes → ~31% attrition. OverTime = No → ~10% attrition. 3x difference — overtime is the single strongest predictor in the dataset.

```
SELECT OverTime,
       COUNT(*) AS headcount,
       ROUND(100.0 * SUM(CASE WHEN Attrition='Yes' THEN 1 ELSE 0 END)
            / COUNT(*), 1) AS attrition_rate_pct
FROM hr_attrition
GROUP BY OverTime;
```

ANSWER — Q15

Male attrition ≈ 17%, Female ≈ 15% (similar). Single employees leave at ~25% vs Married ~12% vs Divorced ~10% — marital status is a meaningful differentiator.

-- Gender

```
SELECT Gender,
       ROUND(100.0 * SUM(CASE WHEN Attrition='Yes' THEN 1 ELSE 0 END)
            / COUNT(*), 1) AS attrition_rate_pct
```

```

FROM hr_attrition GROUP BY Gender;

-- Marital Status
SELECT MaritalStatus,
       ROUND(100.0 * SUM(CASE WHEN Attrition='Yes' THEN 1 ELSE 0 END)
            / COUNT(*), 1) AS attrition_rate_pct
FROM hr_attrition GROUP BY MaritalStatus
ORDER BY attrition_rate_pct DESC;

```

ANSWER — Q16

Leavers earn ~\$4,787/mo on average vs stayers at ~\$6,833/mo — a 30% gap. The gap holds across all departments.

```

WITH leavers AS (
    SELECT Department,
           AVG(MonthlyIncome) AS avg_income_leavers
    FROM hr_attrition WHERE Attrition = 'Yes'
    GROUP BY Department
),
stayers AS (
    SELECT Department,
           AVG(MonthlyIncome) AS avg_income_stayers
    FROM hr_attrition WHERE Attrition = 'No'
    GROUP BY Department
)
SELECT s.Department,
       ROUND(l.avg_income_leavers, 0) AS leavers_income,
       ROUND(s.avg_income_stayers, 0) AS stayers_income
FROM stayers s
JOIN leavers l ON s.Department = l.Department;

```

ANSWER — Q17

Employees not promoted in 4+ years leave at ~18–20% vs ~13% for recently promoted. The signal is real but not as strong as overtime.

```

SELECT
    CASE
        WHEN YearsSinceLastPromotion BETWEEN 0 AND 1 THEN '0-1 years'
        WHEN YearsSinceLastPromotion BETWEEN 2 AND 3 THEN '2-3 years'
        WHEN YearsSinceLastPromotion BETWEEN 4 AND 6 THEN '4-6 years'
        ELSE '7+ years'
    END AS promo_bucket,
    ROUND(100.0 * SUM(CASE WHEN Attrition='Yes' THEN 1 ELSE 0 END)
        / COUNT(*), 1) AS attrition_rate_pct
FROM hr_attrition
GROUP BY promo_bucket;

```

ANSWER — Q18

Interestingly, salary hike % shows weak correlation with attrition in this dataset. Most employees are in the 11–15% range regardless of whether they left.

```
SELECT
  CASE
    WHEN PercentSalaryHike <= 12 THEN 'Low (<=12%)'
    WHEN PercentSalaryHike <= 19 THEN 'Medium (13-19%)'
    ELSE 'High (>=20%)'
  END AS hike_bucket,
  ROUND(100.0 * SUM(CASE WHEN Attrition='Yes' THEN 1 ELSE 0 END)
    / COUNT(*), 1) AS attrition_rate_pct
FROM hr_attrition
GROUP BY hike_bucket;
```

ANSWER — Q19

Leavers score lower on every dimension. Biggest gap: JobSatisfaction (leavers $\approx$ 2.47 vs stayers $\approx$ 2.78) and WorkLifeBalance (leavers $\approx$ 2.66 vs stayers $\approx$ 2.77).

```
SELECT
  ROUND(AVG(CASE WHEN Attrition='Yes' THEN JobSatisfaction END), 2) AS js_leavers,
  ROUND(AVG(CASE WHEN Attrition='No' THEN JobSatisfaction END), 2) AS js_stayers,
  ROUND(AVG(CASE WHEN Attrition='Yes' THEN EnvironmentSatisfaction END), 2) AS es_leavers,
  ROUND(AVG(CASE WHEN Attrition='No' THEN EnvironmentSatisfaction END), 2) AS es_stayers,
  ROUND(AVG(CASE WHEN Attrition='Yes' THEN WorkLifeBalance END), 2) AS wlb_leavers,
  ROUND(AVG(CASE WHEN Attrition='No' THEN WorkLifeBalance END), 2) AS wlb_stayers,
  ROUND(AVG(CASE WHEN Attrition='Yes' THEN RelationshipSatisfaction END), 2) AS rs_leavers,
  ROUND(AVG(CASE WHEN Attrition='No' THEN RelationshipSatisfaction END), 2) AS rs_stayers
FROM hr_attrition;
```

ANSWER — Q20

Level 1 (Low) has the highest attrition at ~22%. Level 4 (Very High) is lowest at ~11%.

```
SELECT
  CASE JobSatisfaction
    WHEN 1 THEN '1-Low' WHEN 2 THEN '2-Medium'
    WHEN 3 THEN '3-High' WHEN 4 THEN '4-Very High'
  END AS satisfaction_label,
  ROUND(100.0 * SUM(CASE WHEN Attrition='Yes' THEN 1 ELSE 0 END)
    / COUNT(*), 1) AS attrition_rate_pct
FROM hr_attrition
GROUP BY JobSatisfaction
ORDER BY JobSatisfaction;
```

ANSWER — Q21

WorkLifeBalance = 1 (Bad): ~31% attrition. Level 2: ~15%. Level 3: ~15%. Level 4 (Best): ~17%. The signal is mainly at the extreme low end.

```
SELECT WorkLifeBalance,
  ROUND(100.0 * SUM(CASE WHEN Attrition='Yes' THEN 1 ELSE 0 END)
```

```
        / COUNT(*), 1) AS attrition_rate_pct  
FROM hr_attrition  
GROUP BY WorkLifeBalance  
ORDER BY WorkLifeBalance;
```

PHASE 3

Advanced SQL — Answers

ANSWER — Q22

Sales Representative ranks #1 (~40%), followed by Human Resources (~23%) and Laboratory Technician (~24%).

```
WITH role_stats AS (  
    SELECT JobRole,  
           ROUND(100.0 * SUM(CASE WHEN Attrition='Yes' THEN 1 ELSE 0 END)  
                 / COUNT(*), 1) AS attrition_rate_pct  
    FROM hr_attrition  
    GROUP BY JobRole  
)  
SELECT JobRole, attrition_rate_pct,  
       RANK() OVER (ORDER BY attrition_rate_pct DESC) AS attrition_rank  
FROM role_stats;
```

ANSWER — Q23

For each department, the top role by attrition: Sales → Sales Representative, R&D; → Laboratory Technician, HR → Human Resources.

```
WITH role_rates AS (  
    SELECT Department, JobRole,  
           ROUND(100.0 * SUM(CASE WHEN Attrition='Yes' THEN 1 ELSE 0 END)  
                 / COUNT(*), 1) AS attrition_rate_pct  
    FROM hr_attrition  
    GROUP BY Department, JobRole  
)  
ranked AS (  
    SELECT *,  
           RANK() OVER (PARTITION BY Department  
                        ORDER BY attrition_rate_pct DESC) AS rnk  
    FROM role_rates  
)  
SELECT Department, JobRole, attrition_rate_pct  
FROM ranked WHERE rnk = 1;
```

ANSWER — Q24

The largest jump in cumulative attrition occurs in years 0–3, confirming early tenure is the highest-risk window.

```
WITH yearly AS (  
    SELECT YearsAtCompany,  
           SUM(CASE WHEN Attrition='Yes' THEN 1 ELSE 0 END) AS leavers  
    FROM hr_attrition  
    GROUP BY YearsAtCompany  
)
```

```

SELECT YearsAtCompany, leavers,
       SUM(leavers) OVER (
         ORDER BY YearsAtCompany
         ROWS UNBOUNDED PRECEDING
       ) AS cumulative_attrition
FROM yearly
ORDER BY YearsAtCompany;

```

ANSWER — Q25

Worst combo: OverTime=Yes + JobSatisfaction=1 → attrition rate exceeds 50% in this group.

```

SELECT OverTime,
       CASE JobSatisfaction
         WHEN 1 THEN '1-Low' WHEN 2 THEN '2-Med'
         WHEN 3 THEN '3-High' WHEN 4 THEN '4-VHigh'
       END AS satisfaction,
       COUNT(*) AS headcount,
       ROUND(100.0 * SUM(CASE WHEN Attrition='Yes' THEN 1 ELSE 0 END)
            / COUNT(*), 1) AS attrition_rate_pct
FROM hr_attrition
GROUP BY OverTime, JobSatisfaction
ORDER BY attrition_rate_pct DESC;

```

ANSWER — Q26

Leavers: avg 8.1 TotalWorkingYears, 5.1 YearsAtCompany, 2.9 YearsInCurrentRole. Stayers: 11.9, 7.4, 4.5.
Leavers are younger and leave before becoming tenured.

```

SELECT Attrition,
       ROUND(AVG(TotalWorkingYears), 1) AS avg_total_exp,
       ROUND(AVG(YearsAtCompany), 1) AS avg_tenure,
       ROUND(AVG(YearsInCurrentRole), 1) AS avg_in_role
FROM hr_attrition
GROUP BY Attrition;

```

ANSWER — Q27

First job (0 previous companies): ~19% attrition. Job hoppers (6+): ~17%. Moderate experience (3–5): ~15%.
The pattern is moderate but not linear.

```

SELECT
  CASE
    WHEN NumCompaniesWorked = 0 THEN '0 - First Job'
    WHEN NumCompaniesWorked BETWEEN 1 AND 2 THEN '1-2 Companies'
    WHEN NumCompaniesWorked BETWEEN 3 AND 5 THEN '3-5 Companies'
    ELSE '6+ Companies'
  END AS exp_bucket,
  ROUND(100.0 * SUM(CASE WHEN Attrition='Yes' THEN 1 ELSE 0 END)
        / COUNT(*), 1) AS attrition_rate_pct
FROM hr_attrition

```



```
GROUP BY exp_bucket;
```

ANSWER — Q28

Most employees score 0–2. Only a small group hits 4–5. This CTE forms the basis for the high-risk list in Q29.

```
WITH dept_avg AS (  
    SELECT Department,  
           AVG(MonthlyIncome) AS dept_avg_income  
    FROM hr_attrition GROUP BY Department  
) ,  
risk_flags AS (  
    SELECT h.EmployeeNumber, h.Department, h.JobRole, h.Attrition,  
           (CASE WHEN h.Overtime = 'Yes' THEN 1 ELSE 0 END  
            + CASE WHEN h.JobSatisfaction <= 2 THEN 1 ELSE 0 END  
            + CASE WHEN h.WorkLifeBalance = 1 THEN 1 ELSE 0 END  
            + CASE WHEN h.YearsSinceLastPromotion >= 4 THEN 1 ELSE 0 END  
            + CASE WHEN h.MonthlyIncome < d.dept_avg_income THEN 1 ELSE 0 END  
           ) AS risk_score  
    FROM hr_attrition h  
    JOIN dept_avg d ON h.Department = d.Department  
)  
SELECT risk_score, COUNT(*) AS employee_count  
FROM risk_flags  
GROUP BY risk_score ORDER BY risk_score;
```

ANSWER — Q29

Expect ~50–80 employees at risk score 4–5. These are your priority retention cases.

```
-- Add WHERE risk_score >= 4 to the CTE above:  
SELECT EmployeeNumber, Department, JobRole, risk_score  
FROM risk_flags  
WHERE risk_score >= 4  
ORDER BY risk_score DESC;
```

ANSWER — Q30

Leavers should have a higher average risk score than stayers — validating the model. A gap of 0.5+ points confirms directional accuracy.

```
SELECT Attrition,  
       ROUND(AVG(risk_score), 2) AS avg_risk_score  
FROM risk_flags  
GROUP BY Attrition;
```

PHASE 4

Power BI Data Model — Answers

ANSWER — Q31

Four essential Power Query transformations: (1) Remove zero-variance columns (EmployeeCount, Over18, StandardHours). (2) Set correct data types — Attrition and OverTime as Text, numeric scores as Whole Number. (3) Rename columns for readability (e.g. 'YrsSinceLastPromotion'). (4) Add AttritionFlag column (see Q32).

■ You can also add AgeGroup and TenureGroup here in Power Query instead of DAX — either approach is valid; Power Query is faster at load time.

ANSWER — Q32

AttritionFlag = 1 for Yes, 0 for No. This lets you SUM() to count leavers, use it as a Y-axis value, and apply CALCULATE filters cleanly.

-- DAX Calculated Column:

```
AttritionFlag = IF([Attrition] = "Yes", 1, 0)
```

ANSWER — Q33

Use SWITCH(TRUE()) for cleaner readability over nested IFs.

```
AgeGroup =  
SWITCH(TRUE(),  
    [Age] < 30,      "Under 30",  
    [Age] < 40,      "30-39",  
    [Age] < 50,      "40-49",  
    "50+"  
)
```

ANSWER — Q34

Same SWITCH pattern for tenure.

```
TenureGroup =  
SWITCH(TRUE(),  
    [YearsAtCompany] <= 2,  "0-2 Years",  
    [YearsAtCompany] <= 5,  "3-5 Years",  
    [YearsAtCompany] <= 10, "6-10 Years",  
    "10+ Years"  
)
```

ANSWER — Q35

No — a Date table is not needed here. There are no date columns in the dataset. YearsAtCompany, TotalWorkingYears etc. are integer durations, not dates. If you had hire dates or exit dates, a Date table would be essential for time intelligence (YoY, rolling 12-month attrition).

■ This question is deliberately tricky — it tests whether you apply tools thoughtfully rather than by default.

PHASE 5

DAX Measures — Answers

ANSWER — Q36

Three foundational measures. All other measures in the report build on these.

```
Total Employees =  
COUNTROWS(hr_attrition)  
  
Total Attrition =  
SUM(hr_attrition[AttritionFlag])  
  
Attrition Rate % =  
DIVIDE([Total Attrition], [Total Employees], 0)
```

ANSWER — Q37

CALCULATE with a filter overrides the current filter context.

```
Avg Monthly Income =  
AVERAGE(hr_attrition[MonthlyIncome])  
  
Avg Income - Leavers =  
CALCULATE(  
    AVERAGE(hr_attrition[MonthlyIncome]),  
    hr_attrition[Attrition] = "Yes"  
)
```

ANSWER — Q38

Write both measures to enable the leavers vs stayers comparison chart.

```
Avg JobSatisfaction - Leavers =  
CALCULATE(  
    AVERAGE(hr_attrition[JobSatisfaction]),  
    hr_attrition[Attrition] = "Yes"  
)  
  
Avg JobSatisfaction - Stayers =  
CALCULATE(  
    AVERAGE(hr_attrition[JobSatisfaction]),  
    hr_attrition[Attrition] = "No"  
)
```

ANSWER — Q39

ALL() removes the current filter context so the denominator is always the full dataset.

```
% of Total Employees =  
DIVIDE(  
    [Total Employees],  
    CALCULATE([Total Employees], ALL(hr_attrition)),  
    0  
)
```

ANSWER — Q40

Filter on two conditions simultaneously inside CALCULATE.

```
High Risk Employee Count =  
CALCULATE(  
    COUNTROWS(hr_attrition),  
    hr_attrition[OverTime] = "Yes",  
    hr_attrition[JobSatisfaction] <= 2  
)
```

ANSWER — Q41

No extra filter needed — slicer context flows into CALCULATE automatically. The measure is context-aware by default.

```
Attrition Rate - Selected Dept =  
CALCULATE([Attrition Rate %])  
  
-- Or just use [Attrition Rate %] directly on the visual.  
-- It responds to slicer context without modification.
```

PHASE 6

Dashboard Build — Answers

ANSWER — Q42

Format: Percentage with 1 decimal. Conditional formatting: use 'Rules' formatting on the card — if value > 0.15, color the callout value Red.

■ In Power BI, go to Format pane → Callout value → Conditional formatting → Rules.

ANSWER — Q43

Legend = Attrition field, Values = Total Employees measure. In Format pane → Detail labels → Label type → Percent of total.

■ Avoid using raw counts on the donut — percentages are more meaningful for executives.

ANSWER — Q44

Sort by the measure: click the '...' menu on the visual → Sort by → Attrition Rate %. The immediate insight is that HR and Sales outpace R&D; significantly.

■ Always sort bar charts by the metric, never alphabetically, unless the chart is a lookup.

ANSWER — Q45

Rate shows the severity per person in a group. The stacked bar shows the absolute volume of attrition. A department with 50% rate but 10 people is less impactful than one with 20% rate and 500 people. Both charts together give the full picture.

■ This dual-chart logic is a standard data storytelling technique interviewers love.

ANSWER — Q46

For 2–4 values: use Tile or Button slicer style (Format → Slicer settings → Style → Tile). For many values (JobRole): use Dropdown. Tile slicers are more touch-friendly and take up less vertical space than a list.

ANSWER — Q47

Expected pattern: leavers (colored red/orange) cluster in the bottom-left quadrant (lower income, fewer total working years). Stayers cluster to the right and upward.

■ In Power BI scatter plots, put the Legend field (Attrition) in the Legend well, not the Size well.

ANSWER — Q48

Sales Representative has the highest rate (~40%). To add data labels: Format pane → Data labels → On → set label position to 'Outside end'.

■ Turn off the Y-axis label when data labels are on — it removes clutter.

ANSWER — Q49

Matrix: Rows = JobLevel, Columns = OverTime, Values = Attrition Rate %. Conditional formatting: Format → Cell elements → Background color → Diverging scale.

■ The cell at JobLevel=1 + OverTime=Yes is typically the darkest cell in the matrix.

ANSWER — Q50

Use AgeGroup on X-axis, [Total Attrition] on Y-axis with a filter on Attrition='Yes', OR display both [Total Attrition] and [Total Employees] and let the user see proportion. The Under 30 group typically has the highest attrition count relative to its size.

ANSWER — Q51

In Power BI, you cannot natively put four different measure names on the X-axis in one chart. Two approaches: (A) Create a separate summary table with unpivoted rows (Satisfaction Type, Score, Attrition Group) — then use that table on the visual. (B) Use four side-by-side small bar charts, one per satisfaction metric.

■ *Option A is more professional but requires M code in Power Query to unpivot. Option B is faster to build. Either is acceptable.*

ANSWER — Q52

A long 'Yes' segment means a high proportion of employees in that group left. For 100% stacked bars, add Attrition to the Legend and YearsSinceLastPromotion (bucketed) to the Y-axis.

■ *100% stacked bars show proportions, not absolute numbers — always add a tooltip showing the actual count to avoid misleading small-group comparisons.*

ANSWER — Q53

Add AttritionFlag as a visual-level filter set to 0 (current employees only). This prevents already-left employees appearing in a 'who should we focus on' list.

■ *This is a common real-world gotcha — risk lists should only show active employees.*

ANSWER — Q54

OverTime=Yes attrition is ~3x higher than OverTime=No. Use a red/orange color for the 'Yes' bar. The gap is large enough that even non-technical viewers will immediately understand the business implication.

PHASE 7

Insights & Storytelling — Answers

ANSWER — Q55

The typical leaver is a single employee under 35, working in Sales or as a Laboratory Technician, required to work overtime, earning below the department average (~\$4,800/mo vs \$6,800 overall average), with low job satisfaction (score 1–2), poor work-life balance, and no promotion in 4+ years. They typically have fewer than 5 years at the company — consistent with early-tenure regret or unmet expectations.

ANSWER — Q56

Overtime is the strongest single predictor: employees who work overtime leave at ~31% vs ~10% for those who don't — a 3x difference. No other single variable in the dataset produces this magnitude of separation.

ANSWER — Q57

Sales Department should be prioritized. Although HR has a slightly higher attrition rate (~26% vs ~21%), Sales has significantly more headcount (~400+ employees) — meaning the absolute number of employees leaving is larger in Sales. The revenue impact of losing experienced sales staff also tends to be higher and more immediate than other functions.

ANSWER — Q58

(a) Compensation: Employees earning below department average leave at higher rates — conduct a pay equity review and close gaps exceeding 15% below median. (b) Promotion: Employees not promoted in 4+ years show elevated attrition — implement a maximum 3-year review cycle with clear promotion criteria. (c) Overtime: 31% attrition among OT employees — audit which teams are consistently requiring overtime and hire additional headcount in those roles. (d) Satisfaction: Low JobSatisfaction (score 1–2) correlates with ~22% attrition — introduce quarterly anonymous pulse surveys with mandatory action plans for teams below threshold.

ANSWER — Q59

Limitation 1: No time dimension — we don't know when employees left, so we can't measure trends or seasonal patterns. Adding hire date and exit date would allow cohort and survival analysis. Limitation 2: No exit interview data — we can measure correlates of attrition but cannot confirm causation. An employee might leave due to family reasons, not overtime, even if they worked overtime. Qualitative exit data would strengthen conclusions.

ANSWER — Q60

The three most impactful visuals for a non-technical HR manager: (1) The Attrition Rate by Department bar chart — immediate answer to 'how bad is it and where.' (2) The OverTime vs Attrition Rate bar chart — a single, dramatic, actionable number. (3) The High Risk Employee table — names and roles of people who may leave next, turning the analysis into a concrete to-do list.